

DATABASE MANAGEMENT SYSTEMS (CS301)

10 Marks Question Bank

UNIT 1 – Introduction to Databases

1. Explain DBMS architecture and the three-level architecture in detail.
2. Discuss File System versus DBMS with suitable examples.
3. Explain logical and physical data independence.
4. Describe the advantages and applications of DBMS.

UNIT 2 – ER Model & Relational Model

1. Explain ER Model with entities, attributes, and relationships.
2. Draw and explain an ER diagram for a college management system.
3. Discuss different types of keys in the relational model.
4. Explain referential integrity with examples.

UNIT 3 – SQL Queries

1. Explain DDL, DML, DCL, and TCL commands with examples.
2. Discuss different types of joins with SQL queries.
3. Explain nested queries and correlated subqueries with examples.
4. Describe stored procedures and their advantages.

UNIT 4 – Functional Dependencies & Normalization

1. Explain functional dependencies and their types with examples.
2. Discuss Armstrong's Axioms and attribute closure.
3. Normalize a relation up to 3NF with suitable examples.
4. Explain BCNF and compare it with 3NF.

UNIT 5 – Transactions & Concurrency Control

1. Explain ACID properties with suitable examples.
2. Discuss concurrency control and locking mechanisms.
3. Explain Two-Phase Locking (2PL) protocol.
4. Describe deadlocks, prevention, avoidance, and recovery techniques.

Exam Preparation Tips

Focus on ER diagrams, SQL query writing, normalization problems, functional dependency analysis, ACID properties, and concurrency control concepts. These are among the most frequently asked 10-mark university examination topics.